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Comparison of the Finite Element and Meshless Methods for Femur Bones of Different Lengths

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Abstract
Variations in femoral length can alter internal stress and strain distributions when subjected to physiological loading. This work compares two femur models with different lengths – 413 mm and 461 mm – subjected to 3000 N and 7000 N forces, using both Finite Element Method and Radial Point Interpolation Method. Both loading scenarios reproduced joint forces typically observed in daily and high-impact activities. Results revealed that the shorter femur experienced higher von Mises stress value (approximately, 103 MPa) and equivalent strain (up to 0.0054), while the longer model reached lower values (79.8 MPa and 0.0042). RPIM produced smoother stress and strain fields compared to FEM, owing to its meshless interpolation, while maintaining consistent mechanical trends. These findings emphasize the relevance of anatomical variation and support meshless approaches in modeling complex biological structures.

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1. Introduction

Does the length of a bone influence how it reacts under stress? This question is highly relevant in the context of computational biomechanics, where simulations are used to evaluate the mechanical performance of bones under physiological loads.

Bone is a living, dynamic, and anisotropic tissue that adapts continuously to its mechanical environment via a process known as bone remodeling. This process involves the formation and resorption of bone tissue in response to mechanical stimulation, a concept rooted in the seminal work of Wolff who first proposed that bone architecture adapts to the loads it experiences [1], [2]. Structurally, bone is a composite material with hierarchical organization designed to provide both mechanical strength and biological stability and support. The femur as the largest and strongest bone in the human body plays a key role in weight-bearing and locomotion, being the prime subject for biomechanical analysis due to its clinical relevance in fractures, implant design and gait dynamics [3], [4]. Its geometry can be significantly different between individuals due to factors such as age, sex, ethnicity, pathology, etc. [5]. These anatomical differences may affect the distribution of stress and strain under load conditions during daily activities [3].

Computational simulations acquire great importance as they investigate biomechanical behavior. To study these biomechanical responses, the Finite Element Method (FEM) has been widely applied due to its accuracy and flexibility in modeling complex biological structures [6]. However, FEM depends heavily on mesh quality and becomes computationally demanding when modeling evolving geometries like in bone remodeling. In contrast, meshless methods such as Radial Point Interpolation Method (RPIM) that bypass mesh generation by using nodes and radial basis functions to interpolate filed variables across the domain, are being developed as competitive alternatives. This method provides greater flexibility and accuracy, offering advantages in cases of geometric complexity and large deformations or patient-specific anatomies from medical imaging [2], [7].

Recent studies have shown that RPIM and related meshless methods produce smoother and realistic representations of trabecular architecture in femoral simulations when compared to FEM, especially under bone remodeling algorithms based on strain energy density minimization [2]. However, direct comparisons between these approaches in the context of bones with varying anatomical features, such as femur length, remain insufficient.

The present study addresses this gap by comparing FEM and RPIM in their ability to simulate the mechanical behavior of two femurs of different lengths subjected to identical physiological loading conditions. Through the analysis of von Mises stress and equivalent strain distributions, this document evaluates the influence of femur length on mechanical performance, and the robustness of each numerical strategy.

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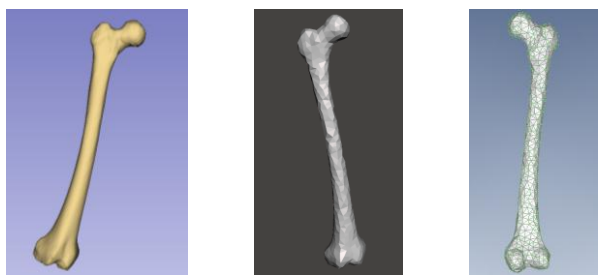
2. Materials and Methods

This section describes the materials and methods used to construct the femur model and perform its analysis.

2.1 Model construction and Material Properties

A CT scan of both legs and pelvis in DICOM format was downloaded from the Embodi3D database. The segmentation of only one of the femur bones was performed in 3D Slicer (Fig.1 (a)) and the resulting 3D model was exported as an STL file, which was then imported into Meshmixer. In this step, the mesh was cleaned using the "Inspector" tool to automatically repair defects. The model was simplified using the "Adaptative Reduce" tool to optimize mesh density. In Meshmixer, the scale and length were adjusted to produce two versions of the same femur with different sizes (413 mm and 461 mm in total length [7]) to simulate two femur bones with different lengths. The result of the steps performed in Meshmixer, for the 413 mm bone, is demonstrated in Fig.1 (b). These models were imported to FEMAP for volumetric meshing using tetrahedral elements (Fig.1 (c)). Two solid meshes were generated with a total of 1324 nodes – 4403 elements and 1320 nodes – 4388 elements, for the femur bone with 413 mm and 461 mm, respectively. After the generation of the mesh, the model was exported as INP file for analysis in FEMAS v2023.1 where the bone was considered as solid isotropic elastic static material to minimize the complexity of the analysis.



(a) (b) (c)

Fig.1 – Construction of the femur model. (a) Segmentation in 3D slicer; (b) Surface mesh in Meshmixer; (c) Tetrahedral mesh in Femap.

In Table 1 we can see the mechanical properties of femur bone [7].

Table 1 – Mechanical properties of femur bone (Adapted from [7]).

Mechanical Properties	Cortical bone
Young's modulus	17000 MPa
Poisson's ratio	0.30
Tensile strength	130 MPa

Only one material was considered in the FEMAS v2023.1 software, defined as solid isotropic. Based on the literature [7], the values used are listed in Table 1.

2.2 Boundary Conditions

To simulate realistic constraint, the only boundary condition applied, in FEMAS v2023.1 software was the fixation of the lateral condyle surface of each femur model. This aimed to replicate the mechanical behavior of a healthy human femur, as illustrated in Fig.2. This boundary condition was established to restrict femur movement in space and is essential for enabling the calculation of stress and strain distributions at each node of the analyzed model.

Fig.2 - Essential boundary conditions applied to the femur bone model, in FEMAS v2023.1.

2.3 Applied Loads

External forces were applied at the femoral head of both bone models to simulate physiological loading. Two load magnitudes were considered: 3000 N and 7000 N, representing typical joint reaction forces observed during daily and high-impact activities, respectively.

The 3000 N load was selected to represent an intensified but physiologically realistic loading scenario, corresponding to approximately 3 to 4.5 times body weight (BW) in an average adult with 70 to 80 kg. This range is supported by in vivo measurements from instrumented hip prostheses during activities such as brisk walking or stair climbing which generate joint contact forces up to 3000 N or more [8], [9].

The 7000 N load was applied to simulate extreme load conditions, such as stumbling, jumping or sudden high-impact movements, where hip joint reaction forces can exceed 6 to 8 times BW. In these scenarios, studies have reported joint contact forces up to approximately 2.5 - 7.3 × BW during stumbling (for example, sudden loss of support) [10], and up to approximately 8.4 × BW during explosive movements such as vertical jumping and push-jerk exercises in trained individuals [11].

The forces were applied at the same mesh nodes located on the articular surface on the femoral head and the direction of the applied force was set normal to the defined boundary surface, as shown in Fig.3.

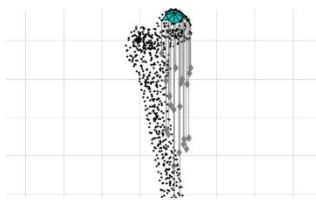
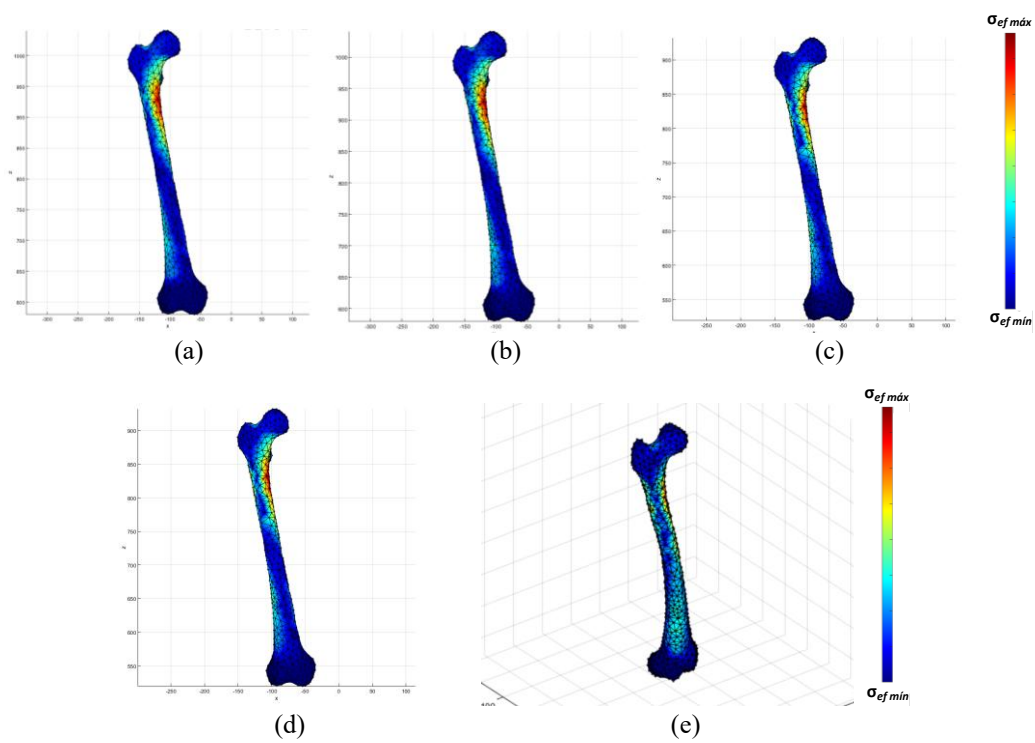
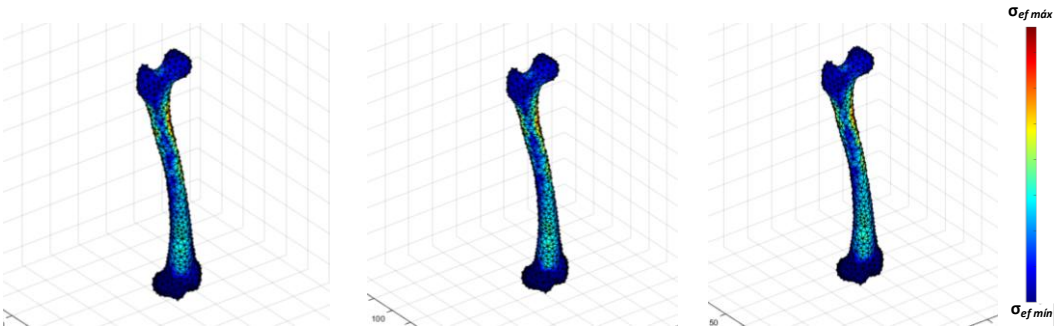


Fig.3 – Location and direction of the applied forces on the femoral head.

3. Results

This section presents the results obtained after applying the forces to both femur models using FEMAS. The primary objective was to evaluate the distribution of effective von Mises stress under two loading scenarios (3000 N and 7000 N). Four cases were analyzed, combining two different femur lengths (413 mm e 461 mm) with two force magnitudes each (3000 N and 7000 N). Both numerical approaches, FEM and RPIM, were applied to each load case to allow comparative evaluation. The resulting stress distributions in Fig.4 illustrate how variations in femur length and load conditions affect internal stress patterns and highlight the differences between FEM (3D) and RPIM (2D).





(f) (g) (h)

Fig.4 – Effective von Mises stress distributions obtained from **RPIM analysis** in (a) 7000 N load applied to the 461 mm femur; (b) 3000 N load applied to the 461 mm femur; (c) 3000 N load applied to the 413 mm femur; (d) 7000 N load applied to the 413 mm femur; and for a **FEM analysis** in (e) 3000 N load applied to the 413 mm femur; (f) 7000 N load applied to the 413 mm femur; (g) 3000 N load applied to the 461 mm femur; (h) 7000 N load applied to the 461 mm femur.

To perform a more detailed analysis, five nodes located in the region where the highest stress concentrations were observed were selected, as shown in Fig. 5, in green.

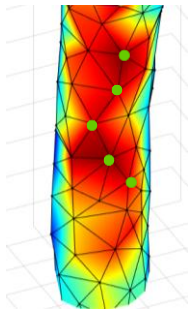
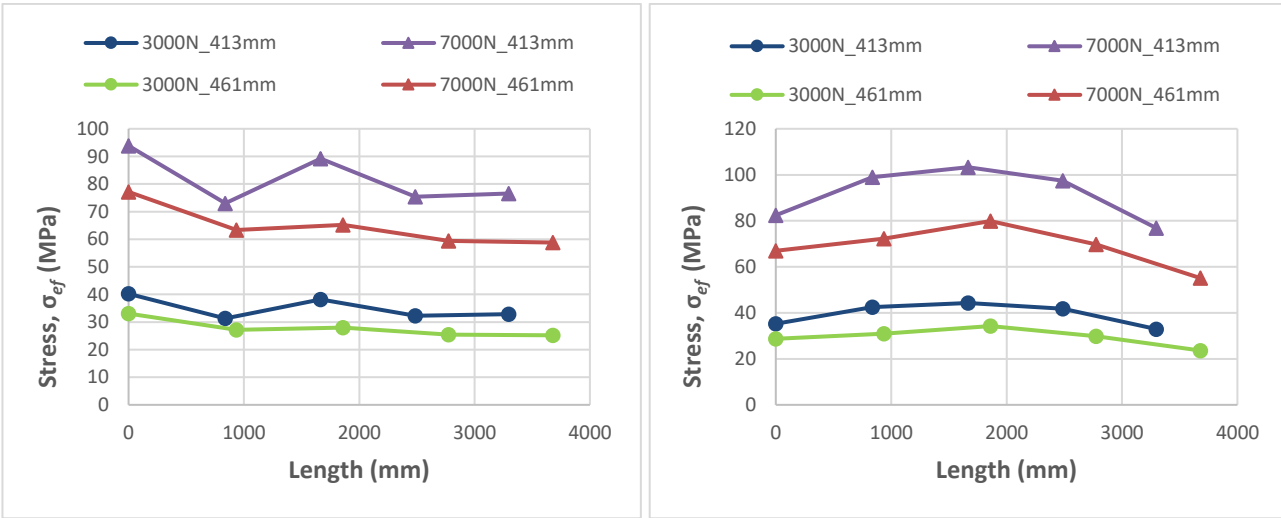
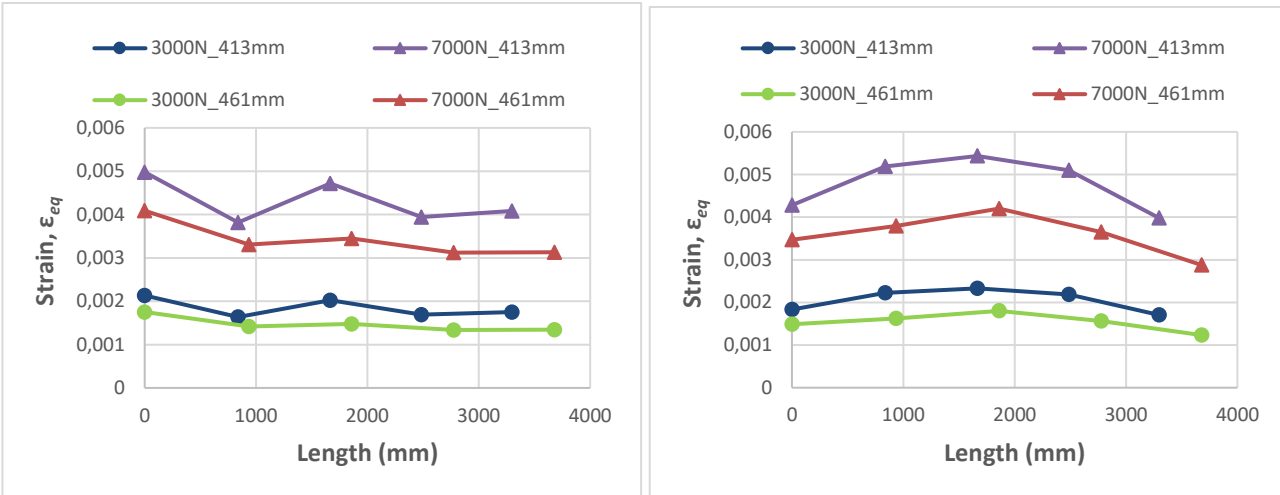


Fig.5 – Points selected in the highest stress region.

These points served as references to evaluate the von Mises effective stress (σ_{ef}) in MPa and the equivalent strain (ε_{eq}) under different loading conditions and analysis methods. Fig.5 presents the graphical distribution of the von Mises stress values obtained through the FEM (a) and RPIM (b) *vs* normalized length in mm, as well as the equivalent strain values computed using FEM (c) and RPIM (d) *vs* normalized length in mm.



(a) (b)



(c) (d)

Fig. 5 – Comparative plots of von Mises effective stress at the selected points obtained using (a) FEM and (b) RPIM methods and equivalent strain distributions for the same points for (c) FEM and (d) RPIM.

4. Discussion

Increasing the applied force from 3000 N to 7000 N results in a consistent rise in both von Mises stress and equivalent strain values across all femur lengths and analysis methods. This trend was expected and reflects the direct relationship between load magnitude and internal mechanical response in bone structures.

The shorter femur (413 mm) showed higher stress and strain levels than the longer one (461 mm) under the same load. For instance, the RPIM simulations under 7000 N showed peak von Mises stress values of approximately 103 MPa for the 413 mm femur, while the 461 mm model reached about 79.8 MPa (Fig.5 (b)). Equivalent strain followed the same trend, with values up to 0.0054 in the shorter bone versus 0.0042 in the longer one (Fig.5 (d)). Under the 3000 N load, the same pattern was observed. The 413 mm femur reached a von Mises stress of approximately 44.2 MPa, while the 461 mm model exhibited a peak value of around 34.2 MPa (Fig.5 (b)). Corresponding equivalent strain values were 0.0023 for the shorter femur and 0.0018 for the longer one (Fig.5 (d)).

A noticeable distinction was observed between the two numerical methods. FEM is based on a structured mesh that discretizes the femur geometry into finite elements. This meshing process can introduce discontinuities or localized variations in the stress or strain results due to element interfaces or irregularities in mesh density. On the other hand, RPIM is a meshless approach, relying instead on scattered nodes to construct the solution. This results in smoother and more continuous stress and strain profiles, as reflected in the more linear and gradual curves observed in the RPIM graphs.

It is important to note that although the color maps shown in Fig.4 provide a helpful visual overview of stress distribution, they do not allow for precise quantitative conclusions. Visually, the distributions appear quite similar across all load cases and femur lengths, with no obvious anomalies. For this reason, a more detailed and localized comparison was analyzed based on point-specific data, as presented in Fig.5.

The graphical comparison of selected points across both methods confirmed this behavior. Despite differences in representation, the general mechanical trends were consistent between FEM and RPIM. RPIM's smooth and linear response is characteristic of meshless methods and reflects a more uniform interpolation of the stress and strain fields. Meanwhile, the FEM results display more localized variations, likely due to the influence of mesh resolution and geometry discretization. Despite these differences in representation, both approaches yield consistent mechanical behavior, reinforcing the validity of RPIM as a reliable and accurate alternative to FEM for biomechanical modeling of complex structures like the femur.

Although no experimental validation was performed, the obtained stress levels agree with values reported in the literature for cortical bone under similar loading conditions, which range from 80 MPa to 130 MPa [12], [13]. This supports the physiological plausibility of the results and further validates the simulation approach.

5. Conclusions

Analyzing femurs of varying lengths under identical physiological loads revealed that bone geometry significantly influences mechanical responses. Shorter femur exhibited increased von Mises stress and strain, indicating a stronger load redistribution over a reduced volume.

The comparison between FEM and RPIM highlighted the strengths of each method. While FEM remains a standard due to its established formulation, RPIM provided smoother, continuous stress and strain fields – particularly advantageous in complex or patient-specific geometries. Despite these differences, both methods were consistent in capturing global mechanical behavior, confirming the reliability of RPIM as an accurate alternative for biomechanical simulations.

This study was limited to linear static conditions and idealizes material assumptions. No experimental validation was performed, and dynamic or adaptive remodeling effects were not considered. Future developments should integrate non-linearities, adaptative remodeling behavior, or experimental validation to enhance physiological accuracy and clinical relevance.

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